Group Member names (must be PRESENT)	What is one topic that is currently difficult to understand? (each person answers)	Do you think Group Assignments are helpful for students next quarter? (each person answers) yes/no
Kate Pun	Difference between network layer and transport layer responsibilities	No
James Anderson	Dijkstra & Bellman Ford	Sort Of
Yash Rao	Network layer forwarding	Yes but fewer is better
Edwin Huang	Link state routing	Kind of
Younwoo Park	Link routing	No

What is the link to your Google doc? CSE 150 Group Asgn 6

Answer the following questions:

A. Longest Prefix Match

This Forwarding Table below has 4 rows (entries) and **all of the prefixes are written in binary.**

Destination Address Range	Output Link Interface
11001000 00010111 00010000 00000000 through 11001000 00010111 00010111 11111111	0
11001000 00010111 00011000 00000000 through 11001000 00010111 00011000 11111111	1
11001000 00010111 00011000 00000000 through 11001000 00010111 00011111 11111111	2
otherwise	3

1. Re-write the forwarding table shown above such that the entries in each row are written in **dotted decimal** notation (32-bit network addresses) in the first column, the mask in the second column and the output port of the router in the third column. Use the table below:

Destination Network Address (dotted decimal)	**Mask** **(CIDR notation)**	**output link interface**
200.23.16.0 through 200.23.23.255	/21	0
200.23.24.0 through 200.23.24.255	/24	1
200.23.24.0 through 200.23.31.255	/21	2
otherwise	-	3

2. What is the purpose of the last row?

The last row serves as the default rule that a router will use to forward a packet out if the packet does not have a destination address that matches any of the specific entries in the forwarding table.

3. How many different IP addresses are represented in the 3rd row?

Subnet mask = 21 bits long

32 - 21 = 11 bits for the host portion

2^{11} = 2048 IP addresses

There are 20248 different IP addresses represented in the third row.

4. Over which interface should an incoming packet with destination address

11001000 00010111 00011000 10111001 be routed?

Destination Address Range	Output Link Interface
11001000 00010111 00010000 00000000 through 11001000 00010111 00010111 11111111	0
11001000 00010111 00011000 00000000 11001000 00010111 00011000 10111001 11001000 00010111 00011000 11111111	1
11001000 00010111 00011000 00000000 through 11001000 00010111 00011111 11111111	2
otherwise	3

The packet should be routed to interface 1.

5. Suppose a packet arrives at the router and its destination IP Address is 200.23.23.5. Over which interface should the router forward this packet?

200.23.23.5 is 11001000 00010111 00010111 00000101

Destination Address Range	Output Link Interface
11001000 00010111 00010000 00000000 11001000 00010111 00010111 00000101 11001000 00010111 00010111 11111111	0
11001000 00010111 00011000 00000000 through 11001000 00010111 00011000 11111111	1
11001000 00010111 00011000 00000000 through 11001000 00010111 00011111 11111111	2
otherwise	3

The packet should be routed to interface 0.

6. Suppose a packet arrives at the router and its destination IP Address is 200.23.24.250. Over which interface should the router forward this packet?

200.23.24.250 is 11001000 00010111 00011000 11111010

Destination Address Range	Output Link Interface
11001000 00010111 00010000 00000000 through 11001000 00010111 00010111 11111111	0
11001000 00010111 00011000 00000000 11001000 00010111 00011000 11111010 11001000 00010111 00011000 11111111	1
11001000 00010111 00011000 00000000 through 11001000 00010111 00011111 11111111	2
otherwise	3

The packet should be routed to interface 1.

7. Suppose a packet arrives at the router and its destination IP Address is 200.26.58.6. Over which interface should the router forward this packet?

200.26.58.6 is 11001000 00011010 00111010 00000110

Destination Address Range	Output Link Interface
11001000 00010111 00010000 00000000 through 11001000 00010111 00010111 11111111	0
11001000 00010111 00011000 00000000 through 11001000 00010111 00011000 11111111	1
11001000 00010111 00011000 00000000 through 11001000 00010111 00011111 11111111	2
11001000 00011010 00111010 00000110	3

The packet should be routed to interface 3.

8. Suppose a packet arrives at the router and its destination IP Address is 200.23.25.40. Over which interface should the router forward this packet?

200.23.25.40 is 11001000 00010111 00011001 00101000

Destination Address Range	Output Link Interface
11001000 00010111 00010000 00000000 through 11001000 00010111 00010111 11111111	0
11001000 00010111 00011000 00000000 through 11001000 00010111 00011000 11111111	1
11001000 00010111 00011000 00000000 11001000 00010111 00011001 00101000 11001000 00010111 00011111 11111111	2
otherwise	3

The packet should be routed to interface 2.

B. Practice with Subnets

1. As a group, solve #4 in Ch4 Practice Problems

i. The network 223.1.0.0/16 can support 256/24 subnets, as 2^8 = 256.

ii. Each /24 subnet can connect to 254 hosts, as 2^8 - 2 = 254 (subtracting the network 223.1.0.0 and broadcast address 223.1.0.255)

iii. Very first subnet: 223.1.0.0/24, Last subnet: 223.1.255.0/24

iv. Range of usable IP addresses 223.1.2.1 - 223.1.2.254, Broadcast address: 223.1.2.255, Maximum hosts in /28 subnets: 14 hosts per subnet.

2. **A harder mask…. Designing Your Network Space**

Suppose your company owns the network space denoted by **189.37.132.0 / 20**

a. Write /20 in binary and dotted decimal
Binary = 1111 1111 . 1111 1111 . 1111 0000 . 0000 0000
Dotted decimal = 255.255.240.0

b. What is the network address (in dotted decimal)? (use the shorthand method on the first 2 bytes) -- show your work

189.37.132.0
255.255.240.0
—----------
189.37.__.0

240 = 1111 (network bits) 0000 (host bits)

132 = 1000 0100

Result = 1000 0000 = 128 in decimal

Network address = 189.37.128.0

c. How many hosts can be on this network? (leave in exponential form)
Number of hosts = 2^12 - 2 = 4094

d. The number of hosts in (c) is large! The company policy has been to assign IP addresses to employees randomly whenever they join the company. This has led to a very unorganized assignment of IP addresses to employees. Suppose your boss wants to organize the IP addresses so that employees/hosts on each floor of your **16-story building** are assigned a continuous address space according to the floor their desk is on. What suggestion could you offer?
We can handle the address space such that each floor can have its own block of addresses. This makes the network more manageable and its IP addresses organized.
If we have 16 floors, then we can have 4096 / 16 = 256 addresses per floor. We have a /20 network, so we can do 4096 / 256 = 16 /24 subnets in our /20 network. Then, we can just assign each subnet to each floor.

Ex)

Floor 1 = 189.37.128.0/24
Floor 2 = 189.37.129.0/24
Floor 3 = 189.37.130.0/24
…
Floor 16 = 189.37.143.0/24

e. You decide to use 4 **bits** from the host portion of the address space to subnet your network. (Why 4?) What is the new mask in dotted decimal and slash notation? Draw out the address space with the rectangle showing the network and host portions like we did in class.
If we use 4 bits from the host portion, then the new mask is 20 + 4 = /24 in slash notation or 255.255.255.0 in dotted decimal. We use 4 bits specifically because it provides 2^4 = 16 possible subnets (one for each floor).

Original /20 Address Space

New /24 Address Space

f. How many subnets can you make with these 4 bits? How many subnets will you assign per floor in the building?

With these 4 bits, 2^4 = 16 subnets can be made. Because we have 16 subnets and 16 floors, each floor can have its own subnet.

g. How many employees can be assigned IP addresses per subnet or floor of the building?

Each subnet has 8 bits reserved for hosts, 2^8 =256 addresses. 256-2 = 254. Each subnet can have 254 employees assigned IP addresses.

h. The first host address in each subnet is assigned to the floor managers. What IP addresses are reserved for managers on the Floors 1, 2 and 16 of the building? (there is no basement, floor 1 is the entry floor)

Floor 1: 223.1.0.1

Floor 2: 223.1.1.1

Floor 16: 223.1.15.1

i. What is the broadcast address on Floors 1, 2 and 16?

Floor 1: 223.1.0.255

Floor 2: 223.1.1.255

Floor 16: 223.1.15.255

j. Does this example help you to understand why subnets are useful for large address spaces?

Yes subnets divide a large address space into smaller, manageable segments which improves efficiency in larger networks.

k. If there was a local router for your company's network with one interface for each floor of the building, how many entries do you need for the router's forwarding table for each floor in this newly subnetted network?

Router's forwarding table requires 16 entries.

C. Border Gateway Protocol (BGP)

Do a bit of reading, discuss (and answer) these questions:

1. How do you rely on DNS?
    a. We rely on DNS to translate domain names like google.com into an IP address. This allows computers to communicate, and prevents us from having to remember numerical IP addresses for websites. We use DNS to browse websites, send emails, and stream content.
2. What is BGP?
    a. BGP is the internet's routing protocol which allows routing information between different networks or autonomous systems. It's how routers decide the best path for data across the internet. It finds the most efficient route by considering multiple paths.
3. How do you rely on BGP?
    a. We rely on BGP everytime we use the internet. It ensures the data finds its way across complex and large networks. BGP routes requests and responses between ISP and destination servers. Cloud providers also rely on BGP and adjust routes when a network fails.
4. Is it true that physical cables carrying your data run from North America to Europe, Asia and South America? Research and check out this map. How do these cables help connect the Internet?
    a. Yes, physical cables carry data across continents. The cables are known as submarine communication cables and they transmit data at extremely high speeds between continents. They offer much higher bandwidth and lower latency than satellite connections. ISPs rely on these cables to exchange traffic with other networks. The dense network of cables ensures redundancy because the data can be rerouted through other cables if one gets damaged.
5. Why is BGP considered to run on an "honor code"? Consider textbook or a research paper Introduction
    a. BGP is considered to run on an "honor code" because it relies on trust between autonomous systems to accurately exchange routing information. It doesn't verify the accuracy of the routes and assumes all peers will share correct routing information. It is impractical to verify all routes because of the vast number of autonomous servers so they must rely on the "honor code" to operate efficiently.

6. Is it possible that a BGP route knows a path, but does not advertise it to peer routers in adjacent Autonomous Systems? Why could this happen? (we will talk about this in class... if we don't get to it, skip this problem)
    a. Yes it is possible that even if a BGP route knows a path it sometimes does not advertise it to a peer router. This could occur for many reasons like, routing policies, a no export community, private routes, or security concerns.
7. In what way is BGP involved in this news story ?
    a. BGP is involved with this news story because when the submarine cables are damaged, traffic must be routed through a different path to maintain internet connection. The large cable disruptions cause BGP to adapt to the loss of connectivity to ensure the internet traffic can bypass the damaged cables. BGP has a critical role in maintaining global internet connectivity even when there are physical damages to the underlying physical cables.